

Highlights

- AW-DPCNN adaptively fuses STFT spectrograms and GADF images via contrast-guided dynamic weighting with dual-channel convolutional stimuli, suppressing destructive interference between heterogeneous representations
- MSCA-VGG16 integrates multi-scale convolution and squeeze-and-excitation channel attention with a compact embedding head for robust multi-granularity fault identification
- Achieves 99.85% accuracy on CWRU 12 kHz drive-end bearing fault benchmark; adaptive weighting contributes 2.93% over fixed fusion baseline
- Demonstrates 97.45% cross-sensor generalization and 94.81% cross-sampling-rate robustness with systematic ablation validation
- Maintains 91.2% accuracy at 5.0 dB SNR under additive Gaussian noise, confirming real-world noise resilience